Savitribai Phule Pune University

T.Y. B.Sc.(Cyber and Digital Science)

SEMESTER- V

(CDS-355)
Lab Course on CDS 352
Cyber Threat Intelligence

Lab Book

Name: ____

College Name:_______________________________________

Roll No:_______________**Division:**__________________________

Academic Year:_____________________________________

CERTIFICATE

This is to certify that Mr./Ms. **:** ____

Seat Number___________of **T.Y.BSC (Cyber and Digital Science) Sem-V**

Has successfully completed Laboratory course (**Cyber Threat Intelligence**) in

The year_________________.

He/ She has scored______________mark out of 10 (For Lab Book).

Subject Teacher **H.O.D./Coordinator**

Internal Examiner **External Examiner**

Introduction

1. **About the workbook**:

   This workbook is intended to be used by T.Y. B.Sc. (Cyber and Digital Science) students for Cyber Threat Intelligence in Sem - V. This workbook is designed by considering all the practical Concepts topics mentioned in the syllabus.

2. **The objectives of this workbook are:**

   1) Defining the scope of the course.
   2) To bring uniformity in the practical conduction and implementation in all colleges affiliated to SPPU.
   3) To have continuous assessment of the course and students.
   4) Providing ready reference for the students during practical implementation.
   5) Provide more options to students so that they can have good practice before facing the examination.
   6) Catering to the demand of slow and fast learners and accordingly providing the practice assignments to them.

3. **How to use this workbook:**

The workbook is divided into 8 assignments.

Editors:

1) Mr. Satyavan Kunjir	Dr. D. Y. Patil ACS College, Pimpri, Pune
2) Mr. Sammed Bukshete	Dr. D. Y. Patil ACS College, Pimpri, Pune
3) Mrs. Renuka Kulkarni	Dr. D. Y. Patil ACS College, Pimpri, Pune
4) Mrs.Lina Chaudhari	Dr. D. Y. Patil ACS College, Pimpri, Pune

Reviewed By:

1. **Dr. Ranjit D. Patil**	Dr. D. Y. Patil ACS College, Pimpri, Pune
2. **Dr. Sujata Patil**	Dr. D. Y. Patil ACS College, Pimpri, Pune

INDEX

Assignment No	Assignment Name	Page Number
1	Threat Intelligence using Search Engines and Google Dork	1
2	Introduction to Threat Analysis using Threatcrowd.org and Netcraft	4
3	Threat Intelligence using Amass and OWASP	7
4	IP and Domain Reputation Check with MX Toolbox, AbuseDb, The Harvester, and Recon-ng	9
5	Introduction to the Dark Web, Dark Web Search, and TOR	15
6	Introduction to OSINT, Shodan, OSTrICa, and Maltego	20
7	Introduction to Email Header Search, Manual Search for Email Header, and Microsoft Email Header Analysis	24
8	Introduction to DNS Information, WHOIS, and Nslookup	28

Signature of Instructor

Assignment 1

Threat Intelligence using Search Engines and Google Dork

Objective:

To explore the concept of threat intelligence using search engines and understand how to use Google Dorks to uncover potential security threats.

Requirements:

- Access to a computer with an internet connection.
- A web browser (preferably Google Chrome).
- Basic knowledge of cyber security and online search techniques.

1. **Threat Intelligence Using Search Engines**

Threat intelligence involves gathering information about potential and existing threats to an organization's security. Search engines can be valuable tools in this process, as they help identify and analyze potential security threats by searching for specific patterns or indicators of compromise. This can include discovering exposed sensitive data, identifying vulnerabilities, or understanding attack methodologies.

2. **Google Dorking**

Google Dorking (also known as Google hacking) uses advanced search operators in Google to find specific information that is not easily accessible through simple queries. It can reveal security vulnerabilities, exposed sensitive information, or misconfigured websites.

- **Google Dork Operators and Their Uses:**

1. **Site**: - Limits the search to a specific domain.
2. **Example**: site: example.com – searches for results only within example.com.

3. **File type**: - Searches for specific types of files.
4. **Example**: filetype: pdf confidential – finds PDF files containing the word "confidential."

5. **Intitle**: - Searches for pages with specific words in the title.
6. **Example**: intitle: “index of" – finds directory listings.

7. **Inurl**: - Searches for URLs containing specific words.
8. **Example**: inurl: admin – finds URLs with "admin" in them.

9. **Intext**: - Searches for specific words in the body of the page.
10. **Example**: intext: “password" – finds pages containing the word "password."

11. **Cache**: - Retrieves the cached version of a page.
12. **Example**: cache: example.com – shows Google’s cached version of the page.

- **Uses of Google Dorking:**

    - Identifying exposed files and directories.
    - Finding sensitive information that has been accidentally published.
    - Locating vulnerabilities in web applications.

Goals:

Slot 1: Threat Intelligence using Search Engines

1. Explain the importance of threat intelligence in cybersecurity.
2. Discuss how search engines can be valuable tools for gathering threat intelligence.
3. Describe the types of information that can be collected using search engines for threat analysis, such as open-source intelligence (OSINT).
4. Provide an overview of the ethical considerations and legal boundaries when conducting threat intelligence research.

Slot 2: Google Dork and Practical Exercise

1. Define what Google Dorking is and its relevance in cybersecurity.
2. Introduce Google Dork operators (e.g., site, intitle, filetype) and explain their uses.
3. Provide a practical exercise for students to conduct threat intelligence research using Google Dorking.
4. For example: Instruct students to find information about a known cybersecurity threat, such as a recent data breach or malware campaign, using Google Dorks.
5. Encourage students to use specific operators to refine their search results.
6. Ask students to document their findings, including any relevant details about the threat, affected organizations, or indicators of compromise (IoCs).

Conclusion:

Summarize the importance of threat intelligence in cybersecurity and the role of search engines, including Google Dorking, in collecting relevant threat information. Highlight the ethical and responsible use of these techniques in cybersecurity research.

Assignment Evaluation

0: Not Done [ ] **1: Incomplete [ ]** **2: Late Complete [ ]**

3: Need Improvement [ ] **4: Complete [ ]** **5: Well Done [ ]**

Signature of Instructor

Assignment 2

Introduction to Threat Analysis using Threatcrowd.org and Netcraft

Objective:

To introduce students to the concept of threat analysis and familiarize them with the use of Threatcrowd.org and Netcraft as tools for gathering threat intelligence.

Requirements:

- Access to a computer with internet connectivity
- A web browser

1. Introduction to threat analysis:

Threat analysis is the process of evaluating and understanding potential threats to an organization's assets, such as data, systems, and personnel. This involves:

1. **Identifying Threats:** Recog

3. nizing possible threats that could exploit vulnerabilities in the system.

4. **Assessing Threats:** Evaluating the likelihood and potential impact of these threats. This can involve analyzing historical data, threat reports, and current trends.

5. **Prioritizing Threats:** Determining which threats pose the most risk based on their likelihood and potential impact.

6. **Mitigation Strategies:** Developing and implementing strategies to mitigate or counteract the identified threats.

7. **Continuous Monitoring:** Regularly updating the threat analysis based on new information and emerging threats.

2. **Threatcroed.org:**

Threatcrowd.org is a threat intelligence platform that aggregates and visualizes threat data.
It allows users to search and analyze threats by domain, IP address, or malware hash.
Key features include:

Visualizations: Provides graphical representations of threat data, helping users understand Relationships and patterns.

Search Capabilities: Allows querying of threat data to identify potential security issues.

Historical Data: Offers historical data on threats, which can be used to track changes over time.

3. **Netcraft:**

Netcraft is a tool for web security analysis that provides:

1. **Web Server Information:** Details about the technologies used by a website, including server Software, Hosting provider, and platform.
2. **Phishing Detection**: Identifies potential phishing sites and provides reports on their activity.
3. **SSL Certificate Analysis:** Checks the validity and configuration of SSL certificates to ensure Secure communications.

Goals:

Slot 1: Introduction to Threat Analysis

1. Explain the significance of threat analysis in cybersecurity and its role in identifying potential security threats.
2. Discuss the key components of threat analysis, such as threat identification, assessment, and mitigation.
3. Introduce students to the concept of threat intelligence and its importance in understanding cyber threats.
4. Provide examples of threats that organizations commonly face, such as malware, phishing, and DDoS attacks.

Slot 2: Using Threatcrowd.org and Netcraft for Threat Analysis

1. Introduce Threatcrowd.org as a threat intelligence platform and Netcraft as a tool for web security analysis.
2. Demonstrate how to access Threatcrowd.org and Netcraft websites in a web browser.
3. Show students how to perform the following tasks using Threatcrowd.org:
4. Search for a domain or IP address to gather threat intelligence.
5. Identify associated threats, malware, and historical data.
6. Explain how to interpret and use the information provided by Threatcrowd.org for threat analysis.
7. Walk students through the use of Netcraft for web server and infrastructure analysis:
8. Show how to enter a website's URL to analyze its web server and hosting information.
9. Explain how to identify potential vulnerabilities or security-related information.
10. Discuss the implications of Netcraft's findings on web security.
11. Encourage students to explore both Threatcrowd.org and Netcraft and practice using these tools for threat analysis.

Conclusion

Summarize the key takeaways from the assignment, emphasizing the importance of threat analysis and the utility of tools like Threatcrowd.org and Netcraft in gathering threat intelligence. Encourage students to continue exploring these tools to enhance their threat analysis skills.

Assignment Evaluation

0: Not Done [ ] **1: Incomplete [ ]** **2: Late Complete [ ]**

3: Need Improvement [ ] **4: Complete [ ]** **5: Well Done [ ]**

Signature of Instructor

Assignment 3

Threat Intelligence using Amass and OWASP

Objective:

To introduce students to threat intelligence using the Amass tool and resources from OWASP (Open Web Application Security Project).

Requirements:

- Access to a computer with an internet connection
- Amass tool (downloadable from https://github.com/OWASP/Amass)
- Basic knowledge of cybersecurity and online search techniques

Threat intelligence using amass & OWASP

1. **Amass Resources**

Amass is an open-source tool for network mapping and attack surface discovery. It helps in Identifying external attack surfaces by discovering subdomains, IP addresses, and other related Data.

- **Steps to Install Amass:**

1. **Prerequisites:** Ensure you have Go (Golang) installed on your computer.
2. **Install Amass**: go install github.com/owasp-amass/amass/v3/...@latest
3. **Verify Installation**: amass –version

2. **OWASP Resources:**

The Open Web Application Security Project (OWASP) provides numerous resources and tools for Improving web security, including:

1. **OWASP ZAP:** A security scanner for finding vulnerabilities in web applications.
2. **OWASP Cheat Sheet Series:** A collection of best practices for various aspects of web security.
3. **OWASP Top Ten:** A list of the top ten security risks to web applications, updated regularly.

Goals:

Slot 1: Introduction to Threat Intelligence using Amass and OWASP

1. Explain the significance of threat intelligence in cybersecurity and its role in identifying potential security threats.
2. Introduce Amass as an open-source tool used for information gathering and threat intelligence.

3. Discuss the capabilities of Amass, including its ability to discover subdomains, domain names, and other critical information.
4. Provide an overview of OWASP (Open Web Application Security Project) and its role in web application security.
5. Explain the resources available from OWASP, such as their web application security guidelines and tools.

Slot 2: Using Amass and OWASP for Threat Intelligence

1. Guide students through the process of downloading and installing Amass on their computers.
2. Demonstrate how to use Amass to perform subdomain discovery on a target domain.
3. Show students how to use specific Amass commands to gather information about subdomains associated with a target domain.
4. Explain how to interpret the results generated by Amass for threat intelligence purposes.
5. Introduce OWASP's resources related to web application security, such as the OWASP Top Ten and the OWASP Web Security Testing Guide.
6. Discuss how to use OWASP's resources to gain insights into common web application security threats and vulnerabilities.
7. Encourage students to explore Amass and OWASP resources and practice using them for threat intelligence gathering and web application security assessment.

Conclusion

Summarize the key takeaways from the assignment, emphasizing the importance of threat intelligence and the utility of tools like Amass and resources from OWASP in gathering threat intelligence and enhancing web application security.

Assignment Evaluation

0: Not Done [ ] **1: Incomplete [ ]** **2: Late Complete [ ]**

3: Need Improvement [ ] **4: Complete [ ]** **5: Well Done [ ]**

Signature of Instructor

Assignment 4

IP and Domain Reputation Check with MX Toolbox, Abuse DB, The Harvester and Recon-ng

Objectives:

To introduce students to IP and domain reputation checks using various tools, including MX Toolbox, AbuseDb, The Harvester, and Recon-ng.

Requirements:

- Access to a computer with internet connectivity
- MX Toolbox (accessible at https://mxtoolbox.com/)
- AbuseDb (accessible at https://www.abuseipdb.com/)
- The Harvester tool (installable from https://github.com/laramies/theHarvester)
- Recon-ng tool (installable from https://github.com/lanmaster53/recon-ng)

1. IP and domain reputation check

- **Significance of IP and Domain Reputation Checks in Cyber security and Threat Analysis**

  IP and Domain Reputation Checks play a crucial role in strengthening cyber security by identifying potentially malicious activity linked to certain IP addresses and domains. These checks are vital components of threat analysis, helping organizations detect, prevent, and mitigate cyber threats.

- **How Reputation is Determined**

  - **Historical Data:** Reputation services rely on historical records of an IP's or domains past behavior. This includes whether it has been used for malicious activities like distributing malware, spamming, or phishing.

  - **Global Threat Intelligence:** Many services aggregate threat data from multiple sources worldwide, analyzing large volumes of data to update reputation scores dynamically.

  - **Analysis of Domain and IP Characteristics:** Factors like domain registration age, SSL certificate validity, geographical location, DNS behavior, and frequency of attacks originating from an IP or domain also influence the reputation score.

- **Examples of Tools for IP and Domain Reputation Checks**

  - **Cisco Talos Intelligence:** Provides threat intelligence services including IP and domain reputation.

- **Virus Total**: Analyzes domains and IP addresses for malicious activity and provides a reputation score based on various sources.

- **Alien Vault OTX:** Offers threat data and reputation information on IP addresses and domains.
- **Spamhaus:** Maintains blacklists of malicious IP addresses and domains often involved in spamming or phishing activities.

2. MX toolbox & Abuse DB

1) **MX Toolbox**

- **Purpose:**

  MX Toolbox is a comprehensive online tool that focuses on email and domain diagnostics. It provides utilities for analyzing DNS records, identifying mail server issues, and checking domain health.

- **Functions:**

  - **DNS Lookup:** Provides details on DNS records like A, MX (mail server), and TXT records.

  - **Blacklist Check:** Checks whether a domain or IP is listed on common email blacklists, indicating possible spam activity.

  - **Email Server Diagnostics:** Verifies whether an email server is correctly configured and operating within best practices, including testing for open relays and email routing problems.

  - **PTR Lookup:** Resolves IP addresses to domain names, useful for checking reverse DNS records.

- **Use in Threat Analysis**

  MX Toolbox helps identify misconfigurations or potential misuse of domain-related services. A domain listed on blacklists could indicate compromised email infrastructure, a common target for phishing attacks.

2) **AbuseDB**

- **Purpose:**

  AbuseDB (Abuse Database) is a collection of IP addresses, domains, and networks known to engage in malicious activities, such as spamming, hacking, and botnet activity.

- **Functions:**

    - **IP and Domain Reputation:** Provides detailed information on the reputation of IP addresses and domains, including whether they are involved in malicious activities like spamming or DDoS attacks.

    - **Reporting Malicious Activity:** Users can report IP addresses that are engaged in abusive behavior, contributing to the database's accuracy.

    - **Blacklisting:** AbuseDB integrates with various cybersecurity tools to provide real-time data on abusive entities, helping organizations block malicious traffic.

- **Use in Threat Analysis:**

    AbuseDB is a critical resource for threat intelligence teams to assess whether a particular IP or domain has a history of malicious activity, helping them decide whether to block or allow traffic.

3. The harvester & Recon-ng

1) **The Harvester**

- **Purpose:**

    The Harvester is an open-source OSINT (Open Source Intelligence) tool used to gather publicly available information (i.e., passive reconnaissance) on a domain, organization, or individual. It is primarily used for gathering data like emails, subdomains, IPs, and employee names.

- **Functions:**

    - **Email Harvesting:** Extracts email addresses associated with a domain from public sources.

    - **Subdomain Enumeration:** Identifies subdomains of a target domain, revealing parts of the organization’s web presence that may not be immediately obvious.

    - **IP Address Gathering:** Collects associated IP addresses of the target, useful for mapping the organization’s network infrastructure.

    - **Search Engines and APIs:** Uses search engines (e.g., Google, Bing, Baidu) and APIs from services like Shodan and VirusTotal to gather intelligence.

- **Use in Threat Analysis:**

    The Harvester is often used during the reconnaissance phase of penetration testing and red teaming, helping security analysts map an organization’s attack surface by identifying potential weak points, such as unprotected subdomains or leaked email addresses.

2) **Recon-ng**

- **Purpose:**

  Recon-ng is a modular OSINT framework similar to Metasploit but designed for reconnaissance. It automates the collection of information about a target from multiple sources and is extensible through its modules.

- **Functions:**
  - **Modules:** Recon-ng offers various modules for different types of reconnaissance, such as domain reconnaissance, email address enumeration, and geolocation data collection.

  - **API Integration:** Supports API keys for external services (like Shodan, Have I Been Pwned, and VirusTotal) to enhance data gathering.

  - **Report Generation:** Provides detailed reports on the collected intelligence, which can be used to visualize the target's exposure or inform further analysis.

  - **Automation:** Recon-ng automates tasks like subdomain discovery, IP geolocation, and SSL certificate analysis, streamlining the intelligence-gathering process.

- **Use in Threat Analysis:**

  Recon-ng enables security analysts to quickly gather and correlate large amounts of data from diverse sources, which helps in identifying potential vulnerabilities and assessing the overall threat landscape of a target organization.

3) How These Tools Work Together in Threat Analysis

1. **Email and Domain Security:**
   - **MX Toolbox** checks the domain's email configuration and ensures that no email misconfigurations or blacklists are contributing to phishing or spam campaigns.
   - **AbuseDB** helps identify whether the domain or associated IPs have been reported for abuse, providing insight into whether it may be involved in malicious activities.

2. **Passive Reconnaissance:**
   - **The Harvester** and **Recon-ng** collect information about the target's online presence, such as emails, subdomains, and IP addresses. This data helps build a profile of the target, including potential entry points for an attacker.

3. **Threat Identification:**
   - **Recon-ng** and **AbuseDB** allow threat intelligence teams to analyze collected data and determine if any IP addresses, domains, or other assets are linked to known malicious activities.

Goals:

Slot 1: Introduction to Threat Analysis

1. Explain the significance of threat analysis in cybersecurity and its role in identifying potential security threats.
2. Discuss the key components of threat analysis, such as threat identification, assessment, and mitigation.
3. Introduce students to the concept of threat intelligence and its importance in understanding cyber threats.
4. Provide examples of threats that organizations commonly face, such as malware, phishing, and DDoS attacks.

Slot 2: Using Threatcrowd.org and Netcraft for Threat Analysis

1. Introduce Threatcrowd.org as a threat intelligence platform and Netcraft as a tool for web security analysis.
2. Demonstrate how to access Threatcrowd.org and Netcraft websites in a web browser.
3. Show students how to perform the following tasks using Threatcrowd.org:
4. Search for a domain or IP address to gather threat intelligence.
5. Identify associated threats, malware, and historical data.
6. Explain how to interpret and use the information provided by Threatcrowd.org for threat analysis.
7. Walk students through the use of Netcraft for web server and infrastructure analysis:
8. Show how to enter a website's URL to analyze its web server and hosting information.
9. Explain how to identify potential vulnerabilities or security-related information.
10. Discuss the implications of Netcraft's findings on web security.
11. Encourage students to explore both Threatcrowd.org and Netcraft and practice using these tools for threat analysis.

Conclusion:

Summarize the key takeaways from the assignment, emphasizing the importance of threat analysis and the utility of tools like Threatcrowd.org and Netcraft in gathering threat intelligence. Encourage students to continue exploring these tools to enhance their threat analysis skills.

Assignment Evaluation

0: Not Done [ ] **1: Incomplete [ ]** **2: Late Complete [ ]**

3: Need Improvement [ ] **4: Complete [ ]** **5: Well Done [ ]**

Signature of Instructor

Assignment 5

Introduction to the Dark Web, Dark Web Search, and TOR

Objectives:

To introduce students to the concept of the Dark Web, provide an overview of Dark Web search techniques, and explain the use of the TOR (The Onion Router) network for accessing the Dark Web.

Requirements:

- Access to a computer with an internet connection
- Web browser (preferably Firefox) and access to the TOR browser (downloadable from https://www.torproject.org/)
- Basic knowledge of internet browsing and online privacy

1. Introduction to dark web

- **Definition:**

  The Dark Web is a portion of the Deep Web that is intentionally hidden and requires special software, configurations, or authorization to access. The most common way to access the Dark Web is through networks like **Tor** (The Onion Router) or **I2P** (Invisible Internet Project).

- **Characteristics:**

  - **Anonymous Access:** The Dark Web allows users to operate anonymously, as it obscures their IP addresses and encrypts their communications, making it difficult to trace their activities.
  - **Not Indexed by Standard Search Engines:** Unlike the Surface Web, the Dark Web is not indexed by traditional search engines. Search engines cannot crawl it, and you need specific URLs and access methods to reach websites within this hidden part of the web.
  - **Use of Special Software:** Accessing the Dark Web requires specialized tools such as the Tor browser, which routes your connection through multiple layers of encryption to protect your identity.
  - **Content:** The Dark Web hosts a variety of content, ranging from legitimate use cases (whistleblowing platforms, privacy-focused communication) to illegal activities (drug trafficking, weapons sales, cybercrime markets, etc.).
  - **Examples of:** Black market websites (such as Silk Road before it was shut down), forums for hacking services, cryptocurrency marketplaces, and privacy-focused communication channels.

2. Dark Web Search

Dark Web search refers to the methods and tools used to explore and access content on the Dark Web. Because the Dark Web is not indexed by traditional search engines like Google, specialized search engines, directories, and techniques are needed to navigate it. Accessing the Dark Web requires understanding how it operates, the specific tools used to enter it, and the specialized search engines that can help locate content.

- **Techniques for Accessing the Dark Web**

1) **Tor Browser**:
   - **Purpose:**

     The Tor browser is the most commonly used tool for accessing the Dark Web. It routes your internet traffic through a global network of volunteer-operated servers, concealing your identity and enabling access to ".onion" sites, which are part of the Tor network.

   - **Process:**
     - Download and install the Tor browser.
     - Connect to the Tor network.
     - Use ".onion" URLs to navigate Dark Web content.
     - Keep in mind that ".onion" addresses are not standard domain names and can be difficult to find without prior knowledge.
2) **I2P (Invisible Internet Project)**:
   - **Purpose:**

     I2P is another privacy-focused network used for accessing hidden services. It provides anonymous communication and is similar to Tor but focuses more on internal network services.

   - **Process:**
     - Install the I2P software and configure your browser.
     - Use I2P-specific websites called "eepsites" to browse anonymously within the I2P network.
     - I2P is less commonly used than Tor but offers strong anonymity for internal communications.
3) **Tails OS**:
   - **Purpose:**

     Tails is a privacy-focused operating system that you can boot from a USB stick. It comes pre-configured with Tor and other security tools, allowing you to browse the Dark Web safely without leaving any traces on your regular system.

   - **Process:**
     - Install Tails OS on a USB stick.
     - Boot your computer from Tails, which includes the Tor browser and other privacy tools.

- Safely browse Dark Web content without risking exposure to your regular operating system.

3. TOR

Tor (The Onion Router) is a free and open-source software and network that enables anonymous communication on the internet. Its primary goal is to protect users' privacy and anonymity by routing their internet traffic through multiple volunteer-operated servers worldwide. This process makes it difficult for anyone to trace the user's location, identity, or online activity, providing anonymity.

- **Key Features of Tor**
  - **Anonymity:** Tor provides strong anonymity by routing traffic through multiple layers of encryption and relays.
  - **Circumventing Censorship:** Tor allows users to access websites and content that might be blocked or restricted in their country.
  - **Hidden Services (.onion Sites):** Tor can be used to access .onion websites, which are only available within the Tor network and provide an additional layer of privacy.

- **Steps to Install the Tor Browser on Your Computer**

Step 1: Download the Tor Browser

- Go to the official Tor Project website: Open your regular browser and visit https://www.torproject.org/.
- Select your operating system: The website will automatically detect your operating system (Windows, macOS, or Linux) and provide the appropriate download link. Click on the download button to start downloading the installation file.

Step 2: Install the Tor Browser

- Run the downloaded file: After the download is complete, open the installer file. On Windows, this might be a .exe file; on macOS, it would be a .dmg file.
- Follow the installation wizard: The installer will guide you through the installation process. Choose your preferred language and follow the prompts to complete the installation.

Step 3: Launch the Tor Browser

- Open the Tor Browser: Once installed, locate the Tor Browser icon on your desktop or in your applications folder and open it.

- Connect to the Tor network: When you first launch the browser, you'll be prompted to connect to the Tor network. Choose “Connect” to begin accessing the network. If you're in a country with internet censorship, you may need to select the “Configure” option to bypass these restrictions.

Step 4: Start Browsing Anonymously

- Browse the internet securely: Once connected, the Tor Browser will open. You can now browse the internet with enhanced anonymity and privacy.

- Access ".onion" sites: You can also access Dark Web content (e.g., .onion sites) by entering their addresses directly into the Tor Browser's address bar.

Goals:

Slot 1: Introduction to the Dark Web

1. Define the Dark Web and its distinction from the Surface Web (visible web).
2. Explain the anonymity and privacy aspects associated with the Dark Web.
3. Discuss the motivations for individuals and organizations to use the Dark Web.
4. Highlight the potential legal and ethical considerations when accessing or conducting activities on the Dark Web.

Slot 2: Dark Web Search and TOR

1. Introduce Dark Web search techniques and search engines that can be used to access Dark Web content.
2. Explain the importance of using the TOR network for accessing the Dark Web securely and anonymously.
3. Describe what TOR (The Onion Router) is and how it functions to provide anonymity.
4. Demonstrate how to download and install the TOR browser on your computer.
5. Guide students through the process of accessing a Dark Web website using the TOR browser.
6. Discuss the precautions and best practices for maintaining privacy and security while using the TOR network.
7. Mention the ethical considerations when accessing and browsing the Dark Web.

Conclusion:

Summarize the key takeaways from the assignment, emphasizing the need for caution, ethical behavior, and understanding the implications of accessing the Dark Web. Discuss the potential benefits and risks associated with the Dark Web.

Assignment Evaluation

0: Not Done [ ] **1: Incomplete [ ]** **2: Late Complete [ ]**

3: Need Improvement [ ] **4: Complete [ ]** **5: Well Done [ ]**

Signature of Instructor

Assignment 6

Introduction to OSINT, Shodan, OSTrICa, and Maltego

Objectives:

To introduce students to Open Source Intelligence (OSINT), Shodan, OSTrICa (Open-Source Threat Intelligence Collector), and Maltego for threat intelligence and reconnaissance.

Requirements:

- Access to a computer with an internet connection
- Web browser
- OSTrICa (installable from https://github.com/aboul3la/Ostinato)
- Maltego (downloadable from https://www.maltego.com/)

1. **Introduction to OSINT, Shodan**

- **OSINT (Open Source Intelligence)**

Open Source Intelligence (OSINT) refers to the process of collecting and analyzing publicly available information to gather actionable intelligence. This information comes from various "open" sources such as websites, social media platforms, public records, forums, news outlets, and other freely accessible resources. OSINT does not involve hacking or accessing private data but instead relies on information that is legally and openly available to anyone.

- **OSINT's Role in Cybersecurity and Threat Intelligence:**

  - Identifying Threat Actors
  - Vulnerability Discovery
  - Social Engineering Prevention
  - Monitoring Cyber Threats
  - Brand Protection and Risk Management
  - Geopolitical Intelligence and Incident Response

- **OSINT Tools Commonly Used in Cybersecurity:**

Several tools are commonly used for OSINT gathering in the context of cybersecurity and threat intelligence:

  - **Maltego:** A powerful tool for conducting deep-link analysis and visualizing relationships between data.

  - **The Harvester:** Used to gather emails, subdomains, IPs, and open ports from public sources.

  - **Shodan**: A search engine for Internet-connected devices that can reveal exposed and potentially vulnerable systems.

- **SpiderFoot**: An automated OSINT tool for gathering intelligence on domains, IPs, and networks.
- **Recon-ng:** A modular framework for performing reconnaissance and OSINT gathering.
- **Google Dorking:** Using advanced Google search operators to find exposed information, such as unsecured files and sensitive data.

- **Shodan**

Shodan is a specialized search engine that is designed to search for and identify internet-connected devices, also known as **"The Search Engine for the Internet of Things (IoT)."** While traditional search engines like Google or Bing index websites and web pages, Shodan indexes devices such as servers, webcams, routers, security systems, printers, and other IoT devices connected to the internet.

- **What Shodan Does?**

Shodan works by scanning the internet and collecting information about devices that are publicly accessible. This includes details about the device's IP address, open ports, services running on the device, and even specific software or firmware versions. This information is useful for cybersecurity professionals, researchers, and system administrators to identify vulnerable or exposed systems.

- **Use Cases for Shodan:**

  - Vulnerability Identification
  - Monitoring Public Exposure
  - Incident Response
  - Research and Data Collection
  - Hackers and Malicious Actors

2. **OSTrICa open source threat intelligence collector**

OSTrICa is an open-source threat intelligence collector tool designed to gather and consolidate data from various open-source intelligence (OSINT) sources. It automates the process of collecting information related to domains, IP addresses, and other indicators of compromise (IOCs) by querying multiple threat intelligence platforms and public databases. This tool is primarily used by cybersecurity professionals for reconnaissance, threat hunting, and enhancing situational awareness. By centralizing threat intelligence data, OSTrICa helps streamline the process of identifying and assessing potential threats.

3. **Maltego**

Maltego is a powerful open-source intelligence (OSINT) and forensics tool designed for conducting deep data analysis and relationship mapping. It is widely used by cybersecurity professionals, investigators, and researchers to gather and visualize information from various sources, helping to uncover complex connections between people, domains, IP addresses, organizations, and more.

- **Use Cases:**
  - **Threat Intelligence:** Identifying and analyzing cyber threats, including domain and IP reputation.
  - **Forensics and Investigation:** Assisting law enforcement and private investigators in tracing criminal activities, fraud, and cybercrime.
  - **Network Security:** Mapping infrastructure, finding exposed assets, and investigating security incidents.

Goals:

Slot 1: Introduction to OSINT

- Define OSINT (Open Source Intelligence) and its significance in cybersecurity and threat intelligence.
- Explain the sources of open-source intelligence, including publicly available information from websites, social media, forums, and other online platforms.
- Discuss the ethical and legal considerations when conducting OSINT research and reconnaissance.

Slot 2: Shodan, OSTrICa, and Maltego

- Introduce Shodan as a search engine designed for finding devices connected to the internet.
- Explain its features, including the ability to search for specific devices, services, and vulnerabilities.
- Demonstrate how to use Shodan to search for and gather information about specific devices or services.
- Provide examples of searches that can reveal vulnerable devices or exposed services.

- Introduce OSTrICa as an open-source threat intelligence collector tool.
- Discuss its capabilities for gathering information from various sources, including social media, search engines, and threat feeds.
- Walk students through the process of installing and using OSTrICa for gathering threat intelligence.
- Present Maltego as a visual link analysis tool for gathering and visualizing OSINT data.

- Explain how Maltego can be used to create graphs and visualize relationships between different pieces of information.
- Demonstrate how to use Maltego to import data collected from OSTrICa and visualize connections between different data points.
- Highlight the importance of information correlation and analysis in OSINT.

Conclusion:

Summarize the key takeaways from the assignment, emphasizing the value of OSINT, Shodan, OSTrICa, and Maltego in threat intelligence and reconnaissance. Discuss the ethical considerations when conducting OSINT research.

Assignment Evaluation

0: Not Done [ ] **1: Incomplete [ ]** **2: Late Complete [ ]**

3: Need Improvement [ ] **4: Complete [ ]** **5: Well Done [ ]**

Signature of Instructor

Assignment 7

Introduction to Email Header Search, Manual Search for Email Header, and Microsoft Email Header Analysis

Objectives:

To introduce students to the concept of email header search, demonstrate manual methods for examining email headers, and explore Microsoft's email header analysis tool.

Requirements:

- Access to a computer with an internet connection
- Web browser

1. Introduction to email header search

An **email header** is a section of an email that contains metadata about the email's route, sender, and other technical details. The information in the header helps in tracking the email's path from the sender to the recipient and provides key insights for email forensics, spam analysis, and troubleshooting.

- **Purpose of Email Header Search**
  - Tracing Email Origin
  - Understanding Delivery Path
  - Authentication and Security
  - Identifying Malicious Activity

- **Key Components of an Email Header**
  - **From:** Email address of the sender.
  - **To:** Email address of the recipient.
  - **Date:** Date and time the email was sent.
  - **Return-Path:** The email address where bounce-back messages are sent.
  - **Received:** Details of each mail server the email passed through.
  - **Message-ID:** A unique identifier for the email.
  - **X-Originating-IP:** Sometimes shows the IP address of the device from which the email originated.

2. Manual search for email header

A manual search for an email header involves inspecting the raw metadata of an email to extract and analyze details about its origin, route, and authenticity. This process is useful for investigating suspicious emails, troubleshooting delivery issues, and verifying sender legitimacy.

Here's how to manually search and analyze an email header:

- **Access the Email Header:**
  - Open the email in your client (e.g., Gmail, Outlook, Yahoo Mail).
  - Find the option to view the raw or full email header. This is typically found in the email's settings or menu (e.g., "Show original" in Gmail, "Properties" in Outlook).
- **Review Key Components:**
  - Received: Tracks the path of the email through various servers.
  - From/To: Shows the sender and recipient email addresses.
  - Return-Path: Indicates where bounce-back messages are sent.
  - Message-ID: A unique identifier for the email.
  - X-Originating-IP: (If available) Shows the IP address of the sender.
- **Analyze for Issues:**
  - Look for inconsistencies or anomalies, such as mismatched IP addresses, domains, or unusual headers, which may indicate phishing or spoofing.
- **Verify Authentication Results:**
  - Check for SPF, DKIM, and DMARC results to confirm the email's authenticity.

3. Microsoft email header analysis

Analyzing email headers in Microsoft Outlook and Outlook.com involves examining the raw metadata to understand the email's origin, path, and authenticity.

Here's a step-by-step guide:

- **Open the Email Header:**
  - In **Microsoft Outlook**, open the email, click on the **"File"** tab, and select **"Properties"**. The header information will be in the **"Internet headers"** section.

- In **Outlook.com**, open the email, click on the three dots (More actions) and select **"View message source"**.

- **Review Key Components:**
  - **Received:** Shows the email's path through different servers.
  - **From/To:** Indicates sender and recipient email addresses.
  - **Return-Path:** Where bounce-back emails are directed.
  - **Message-ID:** A unique identifier for the email.
  - **Authentication Results:** Details on SPF, DKIM, and DMARC checks.

- **Analyze for Issues:**

Look for discrepancies or unusual information in the headers, such as mismatched sender domains or inconsistent IP addresses, which could indicate phishing or spoofing attempts.

Goals:

Slot 1: Introduction to Email Header Search

1. Explain the importance of email headers in email communication and the role they play in email forensics.
2. Discuss how email headers contain valuable information, including sender, receiver, timestamps, and routing details.
3. Introduce the concept of email header search and its relevance in investigating suspicious emails.

Slot 2: Manual Search for Email Header

1. Demonstrate how to access the email header of an email message in popular email clients (e.g., Gmail, Outlook, Yahoo).
2. Show students how to view the raw email header, which contains detailed information about the email's path and origin.
3. Explain how to interpret key elements within the email header, such as the sender's IP address and email servers involved.
4. Discuss the steps for manually searching online resources (e.g., online header analysis

tools, IP lookup services) to gather additional information about email headers.

<u>Slot 3: Microsoft Email Header Analysis Tool</u>

1. Introduce Microsoft's email header analysis tool, accessible at https://mha.azurewebsites.net/.
2. Explain the features and capabilities of the tool for analyzing email headers.
3. Walk students through the process of using the tool to analyze an email header.
4. Provide an example email header or encourage students to input their own.
5. Interpret the results generated by the tool, including the identification of potential issues or anomalies.
6. Discuss the importance of using tools like Microsoft's email header analysis tool in email forensics and security investigations.

Conclusion:

Summarize the key takeaways from the assignment, emphasizing the significance of email headers in email forensics and the utility of tools like Microsoft's email header analysis tool for investigation.

Assignment Evaluation

0: Not Done [ ] **1: Incomplete [ ]** **2: Late Complete [ ]**

3: Need Improvement [ ] **4: Complete [ ]** **5: Well Done [ ]**

Signature of Instructor

Assignment 8

Introduction to DNS Information, WHOIS, and Nslookup

Objectives:

To introduce students to the concept of DNS (Domain Name System) information, demonstrate how to use WHOIS for domain information lookup, and teach how to use Nslookup for DNS queries.

Requirements:

- Access to a computer with an internet connection
- Basic knowledge of networking and internet domains

1. **Introduction to DNS info , OWHOIS**

- **DNS Information**

Domain Name System (DNS) translates human-readable domain names (e.g., example.com) into machine-readable IP addresses (e.g., 192.0.2.1). It allows users to access websites using easy-to-remember names rather than numerical IPs.

Key Points:

- **Domain Names:** Hierarchical names separated by dots (e.g., www.example.com).
- **DNS Records:** Include A (IP address), MX (mail servers), CNAME (aliases), and more.
- **DNS Zones:** Sections of the DNS namespace managed by specific servers.
- **DNS Queries:** Involve recursive or iterative queries to resolve domain names to IP addresses.

Significance:

- **Website Access:** Essential for navigating the web using domain names.
- **Email Delivery:** MX records route emails to the correct mail servers.
- **Security:** TXT records support security measures like SPF and DKIM.
- **Troubleshooting:** Helps diagnose issues with website access and email.

- **WHOIS**

WHOIS is a protocol used to query databases that store registered domain name information. It provides details about the domain's ownership, registration, and contact information.

- **Key Points:**
  - **Domain Registration Details:** Includes the domain owner's name, organization, contact information, and registration dates.
  - **Registrar Information:** Shows the registrar where the domain is registered.
  - **Name Servers:** Lists the DNS servers associated with the domain.
  - **Expiration Date:** Indicates when the domain registration will expire.
- **Use Cases:**
  - **Verify Domain Ownership:** Confirm the owner of a domain.
  - **Check Expiry Dates:** Monitor when domains are due for renewal.
  - **Investigate Domain History:** Research the background and changes in domain ownership.
- **Tools:**

  WHOIS lookup can be performed using command-line tools like whois, or through web-based services such as WHOIS.net or ICANN WHOIS.

2. **Nslookup**

nslookup is a command-line tool used to query DNS servers for information about domain names and IP addresses.

- **Key Functions:**
  - **Retrieve DNS Records:** Obtain various types of DNS records, such as A (IP address), MX (mail servers), and CNAME (aliases).
  - **Domain Lookup:** Find the IP address associated with a domain name.
  - **Reverse Lookup:** Find the domain name associated with an IP address.

- **Usage:**
  - **Basic Query:** nslookup example.com retrieves the IP address for example.com.
  - **Reverse Lookup:** nslookup 192.0.2.1 retrieves the domain name for the IP address 192.0.2.1.

- **Use Cases:**

Troubleshoot DNS issues, verify DNS configurations, and gather domain information.

Goals:

Slot 1: Introduction to DNS Information

1. Explain the role of the DNS (Domain Name System) in translating human-readable domain names into IP addresses for internet communication.
2. Discuss the importance of DNS information in network troubleshooting, security, and domain management.
3. Introduce the concept of WHOIS and Nslookup as tools for gathering DNS-related information.

Slot 2: WHOIS for Domain Information Lookup

1. Demonstrate how to use the WHOIS database to gather information about a domain name.
2. Explain the purpose of WHOIS in providing details about domain ownership, registration, and contact information.
3. Show students how to perform a WHOIS lookup using various online WHOIS services or command-line tools.
4. Interpret the WHOIS results, including information about the domain registrar, registrant contact details, and domain status.
5. Discuss the applications of WHOIS information in domain management, cybersecurity, and legal matters.

Slot 3: Nslookup for DNS Queries

1. Introduce Nslookup as a command-line tool for querying DNS information.
2. Explain how Nslookup can be used to perform various DNS-related tasks, such as IP address resolution and DNS record lookup.
3. Demonstrate how to open a command prompt (Windows) or terminal (Linux/macOS) and use Nslookup.
4. Walk students through sample Nslookup commands:
5. Resolving a domain name to an IP address (forward lookup).
6. Performing a reverse lookup by entering an IP address to find the associated domain name.
7. Querying specific DNS records (e.g., MX records, TXT records) for a domain.
8. Discuss the practical applications of Nslookup in diagnosing DNS issues and verifying DNS configurations.

Conclusion:

Summarize the key takeaways from the assignment, emphasizing the significance of DNS information, WHOIS, and Nslookup in network management, troubleshooting, and security.

Assignment Evaluation

0: Not Done [ ] **1: Incomplete [ ]** **2: Late Complete [ ]**

3: Need Improvement [ ] **4: Complete [ ]** **5: Well Done [ ]**

Signature of Instructor